

PuppetLine: An Interactive System for Embodied Storytelling with LLM-driven Swarm Robots

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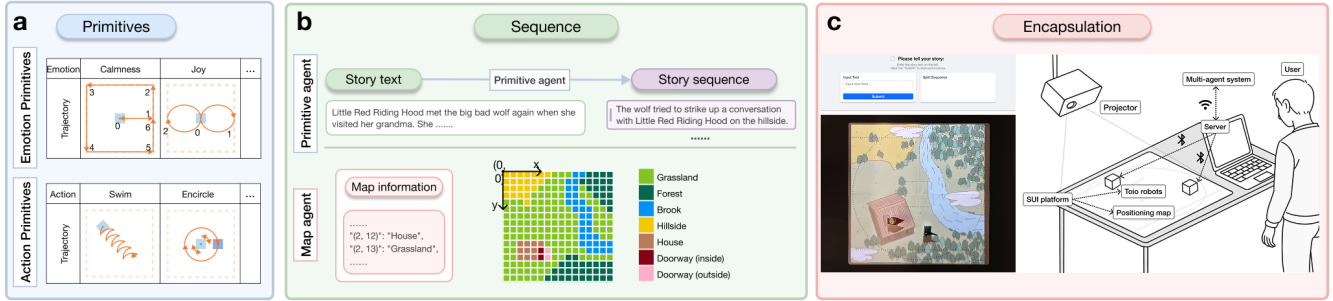


Figure 1: An Interactive System for Embodied Storytelling with LLM-driven Swarm Robots PuppetLine. a) Narrative Primitives. b) Pipeline for LLM-based Storytelling. c) System Design of PuppetLine.

Abstract

Although Swarm User Interfaces (SUIs) offer rich expressive potential for embodied storytelling, their reliance on predefined scripts limits their adaptability to real-time semantic input and open-ended narrative scenarios. In this paper, we present PuppetLine, a storytelling system that integrates large language models (LLMs) with swarm robotics to translate natural language narratives into coordinated multi-robot performances on a tabletop stage. Users input free-form stories through a web interface, which PuppetLine decomposes into structured action sequences using a library of 36 action primitives and 6 emotion primitives. By enabling semantic-to-motion mapping in real time, PuppetLine demonstrates a novel interaction paradigm for embodied storytelling, and opens new possibilities for creative tools at the intersection of natural language processing, robotics, and human-computer interaction.

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CCS Concepts

• **Human-centered computing** → **Interaction devices.**

Keywords

Storytelling; LLM-based Agent; Swarm User Interface; Personalization

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1 Introduction

Swarm User Interfaces (SUIs) are an emerging physical interaction paradigm that leverages the collective motion, shape-changing abilities, and distributed agency of mobile robot swarms to enable novel and expressive tangible interactions. Their spatial and embodied qualities make them particularly suited for storytelling, as they can represent multiple characters, spatial transitions, and narrative tension through coordinated movement [1, 10]. Unlike screen-based

tools, SUIs provide a tangible and performative medium where creators and audiences co-construct stories through physical action. However, existing SUI storytelling systems are often constrained by pre-scripted behaviors and rule-based authoring paradigms [9, 12], requiring designers to manually program motion sequences in advance. This limits scalability and responsiveness, as well as the ability to handle complex or evolving narratives.

To address these challenges, we introduce PuppetLine, a storytelling system that integrates large language models (LLMs) with SUIs to bridge narrative semantics and physical swarm expression. PuppetLine allows users to input stories via a web-based natural language interface, which are interpreted by an LLM and translated into tabletop robot behaviors. By leveraging a library of 36 action primitives and 6 emotion primitives, the system decomposes free-form narrative text into expressive multi-robot performances. This approach lowers the cognitive burden on designers and empowers users without programming expertise to prototype and adapt stories in real time, transforming robot swarms into an accessible medium for embodied storytelling.

2 Related Works

2.1 Creative tools for storytelling and LLM

Storytelling is key in education, creativity, and culture [2, 13]. In HCI, it's seen as an interactive creative process supported by narrative tools [3], though many rely on rigid scripts. Large language models (LLMs) now enable intuitive free-form text input and narrative generation [18], easing creation barriers. Yet, most tools focus on text or visuals, with little work on linking narrative semantics to embodied physical actions.

2.2 Robots for Storytelling

Swarm User Interfaces (SUIs) use coordinated robots for embodied interactions beyond traditional tangible interfaces [8]. Sony's Toio shows strong educational potential. Most robotic storytelling relies on scripted behaviors, limiting flexibility. Modular systems improve reconfigurability but lack semantic understanding [11]. Integrating SUIs with large language models is needed for adaptive, user-driven storytelling.

2.3 Agent for SUI Action Planning

Robotic agents translate high-level commands into low-level actions, exemplified by Skill Transformer [6] and LLM-VLM navigation systems [5, 14]. Swarm planners like Swarm-GPT [7] and Swarm-GPT-Primitive [17] integrate LLMs with motion primitives for better coordination and semantic coherence. However, most focus on functional tasks, with limited exploration in narrative or expressive swarm applications.

This work builds on prior research by creating a narrative motion primitive library for SUI storytelling, enabling LLMs to translate natural language into coordinated multi-robot performances that bridge semantic intent and embodied action.

3 Design and Implementation

3.1 Narrative Primitives Design for Storytelling

To improve SUI expressiveness in storytelling, we designed narrative primitives—modular units based on robot motion primitives—suitable for desktop performances. These planar movements represent core character actions and emotions. Building on Melissa Roemmele et al.'s [15] database of motion patterns, we adapted primitives optimized for Toio robots (Table 1) as core elements in our storytelling system. Through design workshops and iterative testing, we developed a library of emotional primitives grounded in Ekman's six basic emotions and Russell's circumplex model [4, 16], selecting six key emotions: anger, fear, sadness, calmness, joy, and excitement (Figure 2).

Table 1: Single-character and dual-character Actions

Single-character Actions	creep, dance, decelerate, accelerate, go straight, drift, fly, jump, meander, run, spin, stroll, stumble, swim, walk hard, turn, waddle
dual-character Actions	approach, argue with, avoid, capture, chase, encircle, escape, examine, fight, follow, hit, hug, kiss, lead, leave, push, pull, scratch, talk to

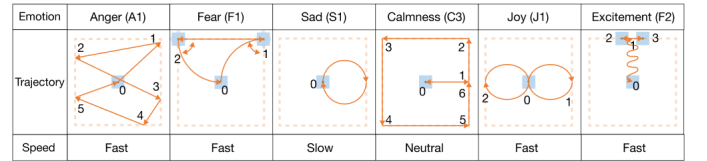


Figure 2: Final Emotional Primitive Design.

3.2 Pipeline Design for LLM-based Storytelling

To convert user stories into coherent, robot-executable sequences on the SUI platform, we employ LLM-based dual-agent for robust semantic understanding. The Primitive-agent extracts key characters and breaks the narrative into simple sentences tied to individual actions or emotions (Figure 1(b)), maintaining character authenticity, plot coherence, and natural language flow. It then assigns motion or emotion primitives to characters for each sentence, creating character-primitive groups that direct robot choreography, defining basic robot motions.

The Map-agent translates narrative primitives into robot motions by interpreting scene maps to assign start and end points for each action. We developed a map with houses, streams, and meadows (Figure 1(b)), using a 16×16 grid of 256 semantically annotated coordinates encoded as text input.

4 Demonstration

To explore real-world applications of embodied storytelling, we developed PuppetLine, an SUI toolkit that integrates a physical performance interface with a cloud-based dual-agent system, enabling

users to create free-form narrative performances directly from text input (Figure 3).

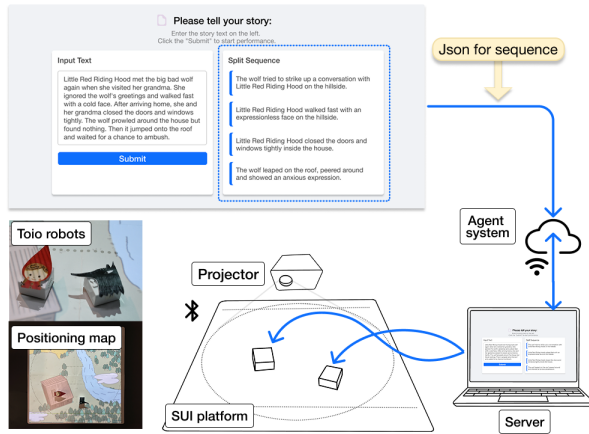


Figure 3: The Hardware and Software System Design of PuppetLine

Through a web interface, users input their story text, which is transmitted to the cloud for semantic processing. The Primitive-agent analyzes the narrative, extracting key characters and mapping their actions or emotions to predefined motion and emotion primitives. The Map-agent then assigns spatial start and end points based on scene map data. Together, these agents generate a JSON-formatted action sequence.

This JSON is parsed by the Toio control program, implemented via Sony Toio’s Python API, to choreograph the robots. Meanwhile, the frontend displays the decomposed scripts and action sequences, allowing users to monitor the breakdown and progression of their story as it unfolds in real time on the tabletop stage.

5 Discussion and Future Works

Compared to traditional narrative tools, PuppetLine’s real-time swarm robot storytelling enhances users’ narrative thinking and serves as a cognitive and creative catalyst in early story development. Beyond individual use, it has potential as a social platform for collaborative storytelling in entertainment, education, and more. PuppetLine thus supports both personal narratives and shared experiences involving co-creation, spatial simulation, and informal learning.

Future work will involve comprehensive user studies to assess PuppetLine’s usability and effectiveness in embodied storytelling with robot swarms. Qualitative interviews will gather insights on semantic-to-motion mapping, creative support, robot expressiveness, and overall user experience to guide system enhancements.

References

- [1] Katharina Buckmayer, Hugo Nicolau, and Ana Cristina Pires. 2024. Storytelling with ALMA: A Multisensory Approach to Enhance Kindergarteners’ Emotion Regulation. In *Proceedings of the 23rd Annual ACM Interaction Design and Children Conference*. 887–891.
- [2] Alejandro Catala, Hannie Gijlers, and Iris Visser. 2023. Guidance in storytelling tables supports emotional development in kindergartners. *Multimedia Tools and Applications* 82, 9 (2023), 12907–12937.
- [3] Alejandro Catala, Mariët Theune, Hannie Gijlers, and Dirk Heylen. 2017. Storytelling as a creative activity in the classroom. In *Proceedings of the 2017 ACM SIGCHI Conference on Creativity and Cognition*. 237–242.
- [4] Paul Ekman. 1992. An argument for basic emotions. *Cognition & emotion* 6, 3–4 (1992), 169–200.
- [5] Wenlong Huang, Chen Wang, Ruohan Zhang, Yunzhu Li, Jiajun Wu, and Li Fei-Fei. 2023. VoxPoser: Composable 3D Value Maps for Robotic Manipulation with Language Models. arXiv:2307.05973 [cs.RO] <https://arxiv.org/abs/2307.05973>
- [6] Xiaoyu Huang, Dhruv Batra, Akshara Rai, and Andrew Szot. 2023. Skill Transformer: A Monolithic Policy for Mobile Manipulation. arXiv:2308.09873 [cs.LG] <https://arxiv.org/abs/2308.09873>
- [7] Aoran Jiao, Tanmay P. Patel, Sanjmi Khurana, Anna-Mariya Korol, Lukas Brunke, Vivek K. Adajania, Utku Culha, Siqi Zhou, and Angela P. Schoellig. 2023. Swarm-GPT: Combining Large Language Models with Safe Motion Planning for Robot Choreography Design. arXiv:2312.01059 [cs.RO] <https://arxiv.org/abs/2312.01059>
- [8] Lawrence H. Kim and Sean Follmer. 2019. SwarmHaptics: Haptic Display with Swarm Robots. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI ’19). Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3290605.3300918
- [9] Iolanda Leite, Marissa McCoy, Monika Lohani, Daniel Ullman, Nicole Salomons, Charlene Stokes, Susan Rivers, and Brian Scassellati. 2015. Emotional storytelling in the classroom: Individual versus group interaction between children and robots. In *Proceedings of the tenth annual ACM/IEEE international conference on human-robot interaction*. 75–82.
- [10] Ting Liu, Francis Quek, and Gabriela Gomez. 2024. Engaging Children in Storytelling Through Tabletop Play: Exploring Construction of Story Ideas through Enactive Actions and Vocalizations. In *Proceedings of the 23rd Annual ACM Interaction Design and Children Conference*. 371–378.
- [11] Ken Nakagaki, Joanne Leong, Jordan L. Tappa, João Wilbert, and Hiroshi Ishii. 2020. HERMITTS: Dynamically Reconfiguring the Interactivity of Self-propelled TUIs with Mechanical Shell Add-ons (UIST ’20). Association for Computing Machinery, New York, NY, USA, 882–896. doi:10.1145/3379337.3415831
- [12] Ken Nakagaki, Jordan L. Tappa, Yi Zheng, Jack Forman, Joanne Leong, Sven Koenig, and Hiroshi Ishii. 2022. (Dis) Appearables: A concept and method for actuated tangible UIs to appear and disappear based on stages. In *Proceedings of the 2022 CHI conference on human factors in computing systems*. 1–13.
- [13] Louise Phillips. 2000. Storytelling: The Seeds of Children’s Creativity. *Australasian Journal of Early Childhood* 25, 3 (2000), 1–5. doi:10.1177/183693910002500302 arXiv:https://doi.org/10.1177/183693910002500302
- [14] Yanyuan Qiao, Yuankai Qi, Zheng Yu, Jing Liu, and Qi Wu. 2023. March in Chat: Interactive Prompting for Remote Embodied Referring Expression. arXiv:2308.10141 [cs.CV] <https://arxiv.org/abs/2308.10141>
- [15] Melissa Roemmele, Soja-Marie Morgens, Andrew S. Gordon, and Louis-Philippe Morency. 2016. Recognizing Human Actions in the Motion Trajectories of Shapes. In *Proceedings of the 21st International Conference on Intelligent User Interfaces* (Sonoma, California, USA) (IUI ’16). Association for Computing Machinery, New York, NY, USA, 271–281. doi:10.1145/2856767.2856793
- [16] James A Russell. 1980. A circumplex model of affect. *Journal of personality and social psychology* 39, 6 (1980), 1161.
- [17] Vedant Vyas, Martin Schuck, Dinushka O. Dahanaggaarachchi, Siqi Zhou, and Angela P. Schoellig. 2024. SwarmGPT-Primitive: A Language-Driven Choreographer for Drone Swarms Using Safe Motion Primitive Composition. arXiv:2412.08428 [cs.RO] <https://arxiv.org/abs/2412.08428>
- [18] Ann Yuan, Andy Coenen, Emily Reif, and Daphne Ippolito. 2022. Wordcraft: story writing with large language models. In *Proceedings of the 27th International Conference on Intelligent User Interfaces*. 841–852.